

MISSION 2

Part I. Problem

Given an Hamiltonian system(in complex form) :

$$A : \dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z}$$

where

$$A := A_1 + iA_2$$

$$B := B_1 + iB_2$$

$$C := C_1 + iC_2$$

Now , find and draw the bifurcation diagram $\Sigma := \Sigma_1 \cup \Sigma_2$

Part II. Solving

Step 1 : to find H

We know that : Without losing generality , We can assume that :

$$Im(C) = 0$$

That is :

$$C_2 = 0$$

Since A is Hamiltonian system , so we have :

$$2A = -\bar{C}$$

$\Rightarrow$

$$\begin{cases} C_2 = 0 \\ A_2 = 0 \\ 2A_1 = -C_1 \end{cases}$$

Therefore , system A is :

$$A : \dot{z} = iz + A_1z^2 + (B_1 + iB_2)\bar{z}^2 - 2A_1z\bar{z}$$

$\Longleftrightarrow$

$$\begin{aligned} \dot{x} + i\dot{y} &= (-y + A_1x^2 - A_1y^2 + B_1x^2 - B_1y^2 + 2B_2xy - 2A_1(x^2 + y^2)) \\ &\quad + i(x + 2xyA_1 - 2xyB_1 + B_2x^2 - B_2y^2) \end{aligned}$$

Therefore , we get the original form of A :

$$A : \begin{cases} \dot{x} = -y + (B_1 - A_1)x^2 + (-B_1 - 3A_1)y^2 + 2B_2xy \\ \dot{y} = x + B_2x^2 - B_2y^2 + 2(A_1 - B_1)xy \end{cases}$$

Again , since A is Hamiltonian , then :

$$\exists H \text{ s.t } \begin{cases} \text{I. } \dot{x} = H_y \\ \text{II. } \dot{y} = -H_x \end{cases}$$

Now , we try to solve the H out :

$\xRightarrow{\text{by I}}$

$$H = \int \dot{x} dy$$

$\Longleftrightarrow$

$$H = \int (-y + (B_1 - A_1)x^2 + (-B_1 - 3A_1)y^2 + 2B_2xy) dy$$

$\Longleftrightarrow$

$$H = -\frac{1}{2}y^2 + (B_1 - A_1)x^2y - \left(\frac{B_1}{3} + A_1\right)y^3 + B_2xy^2 + \text{constant}_1 + f(x)$$

Then , by II :

$\xRightarrow{\text{by II}}$

$$-\dot{y} = H_x = (B_1 - A_1)2xy + B_2y^2 + f'(x)$$

That is :

$$(B_1 - A_1)2xy + B_2y^2 + f'(x) = -x - B_2x^2 + B_2y^2 - 2(A_1 - B_1)xy$$

$\Longleftrightarrow$

$$f'(x) = -x - B_2x^2$$

$\Rightarrow$

$$f(x) = -\frac{x^2}{2} - \frac{B_2}{3}x^3 + \text{constant}_2$$

Therefore :

$$\begin{aligned} H &= -\frac{1}{2}y^2 + (B_1 - A_1)x^2y - \left(\frac{B_1}{3} + A_1\right)y^3 + B_2xy^2 - \frac{x^2}{2} - \frac{B_2}{3}x^3 + \text{constant}_1 + \text{constant}_2 \\ &= -\frac{1}{2}(x^2 + y^2) + (B_1 - A_1)x^2y - \left(\frac{B_1}{3} + A_1\right)y^3 + B_2xy^2 - \frac{B_2}{3}x^3 + \text{constant}_3 \end{aligned}$$

Where $\text{constant}_3 := \text{constant}_1 + \text{constant}_2$

Step 2 : draw the graph of Σ_1

By the definition of Σ_1 :

$$\exists (x_0, y_0) \text{ s.t } \begin{cases} H_x(x_0, y_0, \lambda) = 0 \\ H_y(x_0, y_0, \lambda) = 0 \\ \det(\text{Hess}H(x_0, y_0, \lambda)) = 0 \end{cases}$$

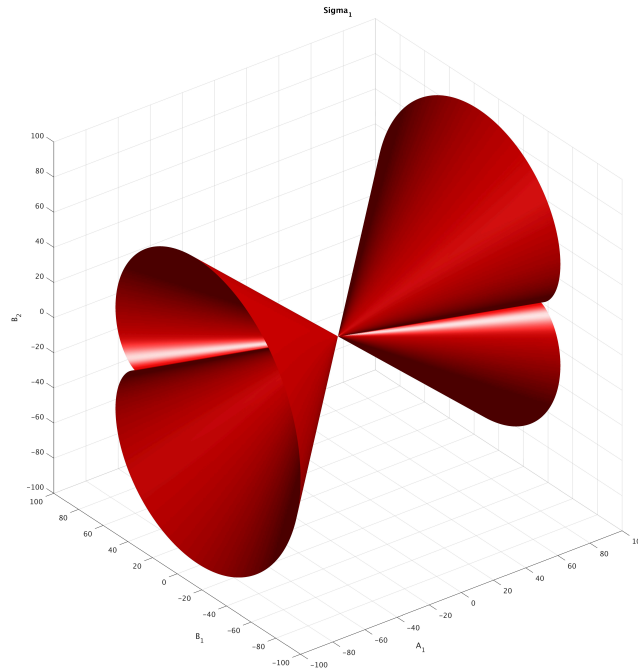
After computing $H_x, H_y, \det(\text{Hess}H)$, we get :

$$\begin{cases} (H_x) : 0 = -x_0 - B_2x_0^2 + B_2y_0^2 - 2(A_1 - B_1)x_0y_0 \\ (H_y) : 0 = -y_0 + (B_1 - A_1)x_0^2 + (-B_1 - 3A_1)y_0^2 + 2B_2x_0y_0 \\ (\det \text{Hess}) : 0 = 1 + 8A_1y_0 - (4B_2^2 + 4(A_1 - B_1)^2)x_0^2 + (12A_1^2 - 4B_1^2 - 4B_2^2 - 8A_1B_1)y_0^2 + 16A_1B_2x_0y_0 \end{cases}$$

Cancel out the x_0, y_0
 $\Rightarrow$

$$A_1^4 + 8A_1^3B_1 + 18A_1^2B_1^2 - 27B_1^4 + 18A_1^2B_2^2 - 54B_1^2B_2^2 - 27B_2^4 = 0 \quad (*)$$

We draw the graph of (*) and get the graph of Σ_1 :



Step 3 : draw the graph of Σ_2

By the definition of Σ_2 :

$$H(x, y, \lambda) \text{ has two different critical points } (x_1, y_1), (x_2, y_2) \text{ s.t. } H(x_1, y_1, \lambda) = H(x_2, y_2, \lambda)$$

That is :

$$\begin{cases} H_x(x_1, y_1, \lambda) = 0 \\ H_x(x_2, y_2, \lambda) = 0 \\ H_y(x_1, y_2, \lambda) = 0 \\ H_y(x_2, y_2, \lambda) = 0 \\ H(x_1, y_1, \lambda) = H(x_2, y_2, \lambda) \end{cases}$$

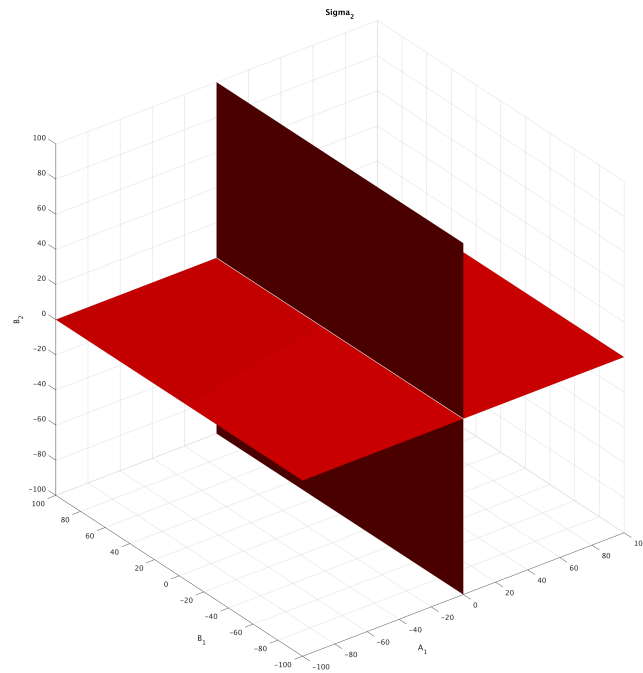
Since we have compute H_x, H_y before , So that is :

$$\begin{cases} (H_x) : 0 = -x_1 - B_2x_1^2 + B_2y_1^2 - 2(A_1 - B_1)x_1y_1 \\ (H_x) : 0 = -x_2 - B_2x_2^2 + B_2y_2^2 - 2(A_1 - B_1)x_2y_2 \\ (H_y) : 0 = -y_1 + (B_1 - A_1)x_1^2 + (-B_1 - 3A_1)y_1^2 + 2B_2x_1y_1 \\ (H_y) : 0 = -y_2 + (B_1 - A_1)x_2^2 + (-B_1 - 3A_1)y_2^2 + 2B_2x_2y_2 \\ (H) : -\frac{1}{2}(x_1^2 + y_1^2) + (B_1 - A_1)x_1^2y_1 - (\frac{B_1}{3} + A_1)y_1^3 + B_2x_1y_1^2 - \frac{B_2}{3}x_1^3 \\ \quad = -\frac{1}{2}(x_2^2 + y_2^2) + (B_1 - A_1)x_2^2y_2 - (\frac{B_1}{3} + A_1)y_2^3 + B_2x_2y_2^2 - \frac{B_2}{3}x_2^3 \end{cases}$$

Cancel out the x_1, y_1, x_2, y_2
 $\Rightarrow$

$$A_1B_2 = 0 \quad (**)$$

We draw the graph of $(**)$ and get the graph of Σ_2 :

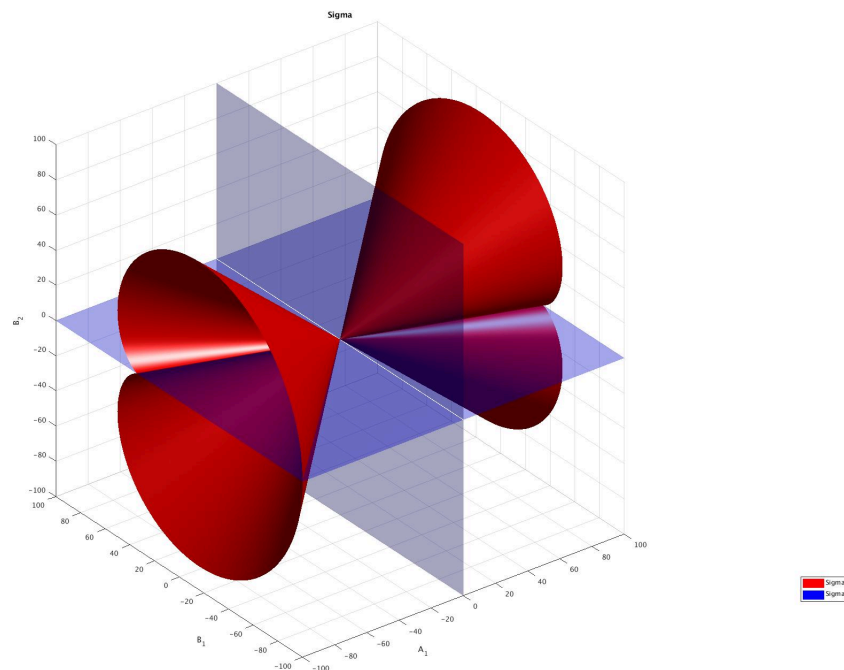


Step 3 : draw the graph of Σ

By the definition of Σ :

$$\Sigma := \Sigma_1 \cup \Sigma_2$$

Therefore , the graph of Σ is the union of Σ_1 (red) and Σ_2 (blue) , that is :



Part III. Remarks

Actually , it is difficult to get the result of (*) and (**)

So , I use the *Mathematica* to solve it and get the result of (*) and (**)

The following are the code of getting (*) and (**):

The code of get (*) :

INPUT :

```
ClearAll["Global`*"]

H[x_, y_, A1_, B1_, B2_] := -1/2 (x^2 + y^2) - 1/3 B2 x^3 + B2 x y^2 + (B1 - A1) x^2 y - (B1 + 3 A1)/3 y^3;

eqs = {
D[H[x, y, A1, B1, B2], x] == 0,
D[H[x, y, A1, B1, B2], y] == 0,
Det[D[H[x, y, A1, B1, B2], {{x, y}, 2}]] == 0
};

GroebnerBasis[eqs, {A1, B1, B2}, {x, y}]
```

OUTPUT :

```
{A1^4 + 8 A1^3 B1 + 18 A1^2 B1^2 - 27 B1^4 + 18 A1^2 B2^2 - 54 B1^2 B2^2 - 27 B2^4}
```

The code of get (**):

INPUT :

```
ClearAll["Global`*"]

H[x_, y_, A1_, B1_, B2_] := -1/2 (x^2 + y^2) - 1/3 B2 x^3 + B2 x y^2 + (B1 - A1) x^2 y - (B1 + 3 A1)/3 y^3;

grad[x_, y_, A1_, B1_, B2_] := {D[H[x, y, A1, B1, B2], x], D[H[x, y, A1, B1, B2], y]};

eqsSigma2 = {grad[x1, y1, A1, B1, B2][[1]] == 0,
grad[x1, y1, A1, B1, B2][[2]] == 0,
grad[x2, y2, A1, B1, B2][[1]] == 0,
grad[x2, y2, A1, B1, B2][[2]] == 0,
H[x1, y1, A1, B1, B2] - H[x2, y2, A1, B1, B2] == 0,
t ((x1 - x2)^2 + (y1 - y2)^2) - 1 == 0};

GroebnerBasis[eqsSigma2, {A1, B1, B2}, {x1, y1, x2, y2, t}]
```

OUTPUT :

```
{A1 B2}
```